

Elastic Net Regression

Elastic Net is a regularized linear regression that combines:

- L1 penalty $\Rightarrow$ sparsity (feature selection)
- L2 penalty (Ridge) $\Rightarrow$ Stability (handles multicollinearity)

$$L = \sum (y_i - \hat{y}_i)^2 + \underline{a \|w\|^2} + \underline{b \|w\|}$$

$a \rightarrow$ Lasso strength

$b \rightarrow$ Ridge strength

$$\lambda = a + b, \quad \lambda_{\text{ratio}} = \frac{a}{a+b}$$

$$\text{if } \lambda_{\text{ratio}} \geq 0.9 \quad \lambda_{\text{ratio}} = \frac{a}{\lambda} \Rightarrow$$

$$a = \lambda \times \lambda_{\text{ratio}}$$

$$b = \lambda - a$$

Ridge (90%) $\rightarrow$ Ridge $\rightarrow$ 10% Lasso

{ default values : $\lambda = 1, \lambda_{\text{ratio}} = 0.5$

When to use Elastic Net:-

- when feature are highly correlated (Multicollinearity):-
 - $\rightarrow$ Lasso $\Rightarrow$ picks one feature & drops others randomly
 - $\rightarrow$ Ridge $\Rightarrow$ keeps all features, no selection $\Rightarrow$ Elastic Net keeps correlated features together (grouping)
- when you want feature selection + stability:-
- when number of features $>$ number of samples
- when you don't know whether L1 or L2 is better

when not to use Elastic Net:-

- $\rightarrow$ very small datasets + few features $\Rightarrow$ simple / Ridge is enough
- $\rightarrow$ when you want hard feature elimination only $\rightarrow$ Lasso
- $\rightarrow$ when all features are equally important $\rightarrow$ Ridge